AIR UNIVERITY

Department of Electrical

College of Engineering

ARTIFICIAL INTELLIGENCE LAB

LAB 05

Lab Objective: Dimensionality Reduction – Principal Component Analysis (PCA)

Class ID: ______________________ **LabDate:**

Student's ID: ___________________

LAB 05	Dimensionality Reduction – Principal Component Analysis (PCA)				
PLO's	PLO3 – Design and Development		Bloom's Taxonomy	C3 – Apply	
LAB TASK PERFORMANCE					
CLO's	**Aspects of Assessments**	**Excellent (75-100%)**	**Average (50-75%)**	**Poor (<50%)**	**M**
CLO2 100%	**Design & Development** Design solution for problem that meets specified needs by using the concept of artificial intelligence.	A complete solution / Explain necessary requirements according to task description with great use of time and resource material.	Solution was complete but need minor modifications /student could have followed specification more closely.	Solution was complete but did not work, needed several modifications / did not make correct use of resource material or instructions.	
				Total Marks: 10	

Graded by Engineer: _________________________________ **Date:** ______________________

Remarks: ____

Objective:

1. Make attendee familiar with the concept of dimensionality reduction
2. Elaborate the concept of Principal Component Analysis (PCA)
3. Hands-on implementation on Python

Dimensionality Reduction

Dimensionality reduction, or **dimension reduction**, is the transformation of data from a high-dimensional space into a low-dimensional space so that the low-dimensional representation retains some meaningful properties of the original data, ideally close to its intrinsic dimension. Working in high-dimensional spaces can be undesirable for many reasons; raw data are often sparse as a consequence of the curse of dimensionality, and analyzing the data is usually computationally intractable. Dimensionality reduction is common in fields that deal with large numbers of observations and/or large numbers of variables, such as signal processing, speech recognition, neuroinformatics, and bioinformatics

Principal Component Analysis

Principal Component Analysis (PCA) is a **linear dimensionality reduction** technique that can be utilized for extracting information from a high-dimensional space by projecting it into a lower-dimensional sub-space. It tries to preserve the essential parts that have more variation of the data and remove the non-essential parts with fewer variation.

Dimensions are nothing but features that represent the data. For example, A 28 X 28 image has 784 picture elements (pixels) that are the dimensions or features which together represent that image.

One important thing to note about PCA is that it is an **Unsupervised** dimensionality reduction technique, you can cluster the similar data points based on the feature correlation between them without any supervision (or labels)

Features / Attributes / Variables

Samples

	sepal-length	sepal-width	petal-length	petal-width
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

But where all you can apply PCA?

1. **Data Visualization**: When working on any data related problem, the challenge in today's world is the sheer volume of data, and the variables/features that define that data. To solve a problem where data is the key, you need extensive data exploration like finding out how the variables are correlated or understanding the distribution of a few variables. Considering that there are a large number of variables or dimensions along which the data is distributed, visualization can be a challenge and almost impossible.
   Hence, PCA can do that for you since it projects the data into a lower dimension, thereby allowing you to visualize the data in a 2D or 3D space with a naked eye.
2. **Speeding Machine Learning (ML) Algorithm**: Since PCA's main idea is dimensionality reduction, you can leverage that to speed up your machine learning algorithm's training and testing time considering your data has a lot of features, and the ML algorithm's learning is too slow.

At an abstract level, you take a dataset having many features, and you simplify that dataset by selecting a few Principal Components from original features.

What is a Principal Component?

Principal components are the key to PCA; they represent what's underneath the hood of your data. In a layman term, when the data is projected into a lower dimension (assume three dimensions) from a higher space, the three dimensions are nothing but the three Principal Components that captures (or holds) most of the variance (information) of your data.

Principal components have both direction and magnitude. The direction represents across which *principal axes* the data is mostly spread out or has most variance and the magnitude signifies the amount of variance that Principal Component captures of the data when projected onto that axis. The principal components are a straight line, and the first principal component holds the most variance in the data. Each subsequent principal component is orthogonal to the last and has a lesser variance. In this way, given a set of *x* correlated variables over *y* samples you achieve a set of *u* uncorrelated principal components over the same *y* samples.

The reason you achieve uncorrelated principal components from the original features is that the correlated features contribute to the same principal component, thereby reducing the original data features into uncorrelated principal components; each representing a different set of correlated features with different amounts of variation.

Each principal component represents a percentage of total variation captured from the data.

Hands-on Activity

In the hands-on activity we will work on the IRIS data set and will use PCA to transform the data to lower dimensional space

```
import numpy as np
import pandas as pd #package to fetch the data

#from google.colab import files # Commented out as not needed in current
execution environment
#uploaded=files.upload() # Uncomment this line to enable file upload
#import io
#df=pd.read_csv(io.BytesIO(uploaded['Iris.csv']),header=None)
# If you are running outside of Google Colab you will need to directly load
your dataframe
# using pd.read_csv() or another method to load data.

# Replace 'Iris.csv' with your actual CSV file path if needed
df = pd.read_csv('Iris.csv', header=0)

# Convert the dataframe to a NumPy array
df1 = df.to_numpy()

# Select columns 1 to 4 (SepalLengthCm, SepalWidthCm, PetalLengthCm,
PetalWidthCm) as features
# Skip the 'Id' column (index 0) and the header row.
X = df1[1:, 1:5]  # Start from the second row (index 1) to exclude the header

# The rest of your code remains the same
from sklearn.preprocessing import StandardScaler
# Standard Scaler normalizes all data to have zero mean and variance 1
X = StandardScaler().fit_transform(X)

from sklearn.decomposition import PCA
```

```
pca = PCA(n_components=2)
transformed_data = pca.fit_transform(X)
transformed_data
```

(Note: You may convert the code for Spyder environment yourself as explained in earlier labs)

Lab Tasks:

(CLO2, Marks 9)

Complete hands-on activities as prescribed above.

Plot scatter plot of original data with different colors for the 3 classes

What is the size of the transformed data? Which two parameters are selected by the algorithm, Give answer with reasoning and justification? If your increase the parameters to 3, then which additional feature is selected? Plot new scatter plots to see if data visualization is improved?

Another alternative way to understand Principle Component Analysis through Eigen Values and Eigen Vectors. Find the Eigen values and Eigen Vectors of the following matrices

A=np.array([[2 ,3, 3],[2,2, -3], [4,-3.1,2]])

A=np.array([[2 ,3, 3],[2,2, 3], [4,3.1,7]])

Comment on the difference in values and relate this to the principle components?

(Hint: You can make use of the command np.linalg.eig(A) for this task)

Is PCA a supervised method or an un supervised method? Give reason for your answer

You are now asked to perform PCA on the IRIS dataset as done in the hands-on activity, but by using the number of principal components which preserve 95% variance in the data. Design an algorithm for this and analyze your observations, how they are different from previous part.

(CLO 6, Marks 1) Perform properly handling of lab.